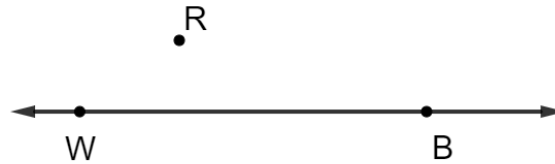


Geometry
Unit 1
Lesson 1 Practice

Name Answer Key
Date _____
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Point R and $\overleftrightarrow{WB}$ are shown.



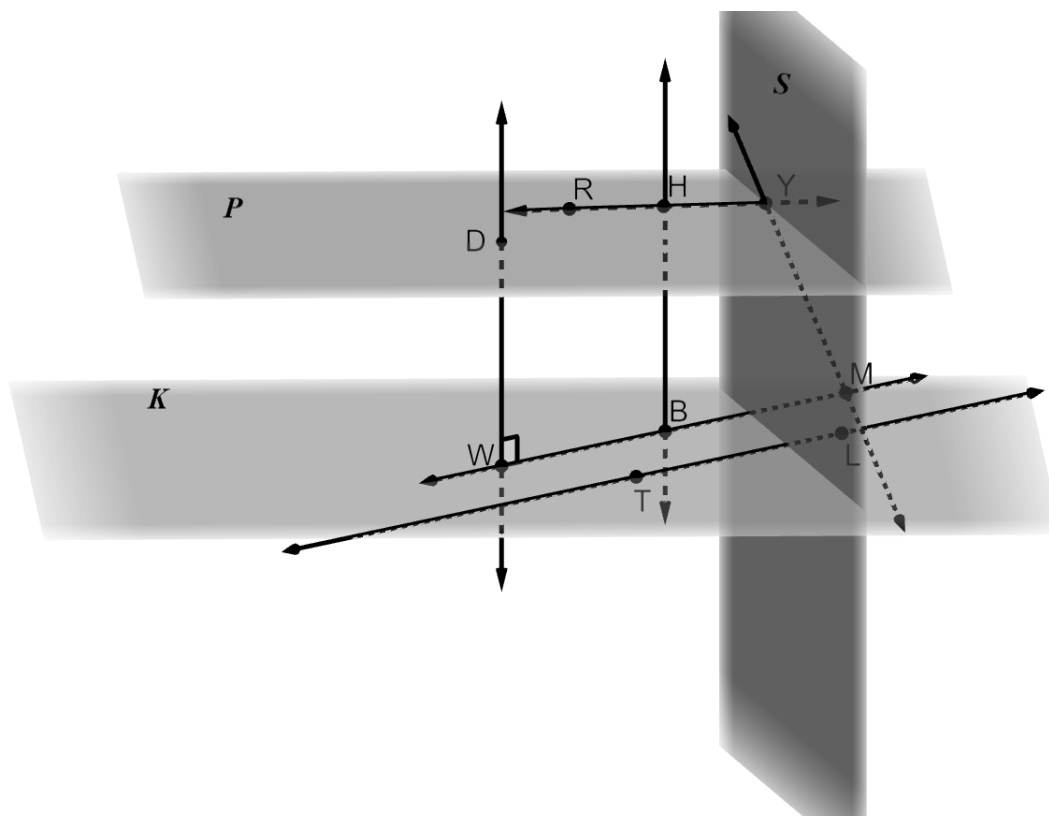
Steps to construct a line that passes through point R and is parallel to $\overleftrightarrow{WB}$ are shown.

Step	Description
A	Fold $\overleftrightarrow{WB}$ so that it coincides with itself and goes through point R .
B	Draw a line on the fold that passes through the dotted line and is parallel to $\overleftrightarrow{WB}$.
C	Fold $\overleftrightarrow{WB}$ so that it coincides with itself and the fold goes through point R .
D	Draw a dotted line through point R and $\overleftrightarrow{WB}$ to represent the fold. Label the point of intersection K .
E	Name the parallel line $\overleftrightarrow{NY}$.

Use the letter of the steps to correctly reorder the steps for constructing a parallel line.

	Letter
1.	C
2.	D
3.	A
4.	B
5.	E

Parallel planes P and K are shown where plane S is perpendicular to both planes P and K .



6. Identify an object, if there is one, that is considered to be one-dimensional, has no height and no width, and has infinite length.

Any of the lines on the image, such as $\overleftrightarrow{RH}$, $\overleftrightarrow{DW}$, $\overleftrightarrow{WM}$, and $\overleftrightarrow{TL}$

7. Identify two lines, if there are any, that are skew lines.

Answers may vary. Possible answer: $\overleftrightarrow{DW}$ and $\overleftrightarrow{MY}$

8. Identify two lines, if there are any, that lie on the same plane.

$\overleftrightarrow{WM}$ and $\overleftrightarrow{TL}$

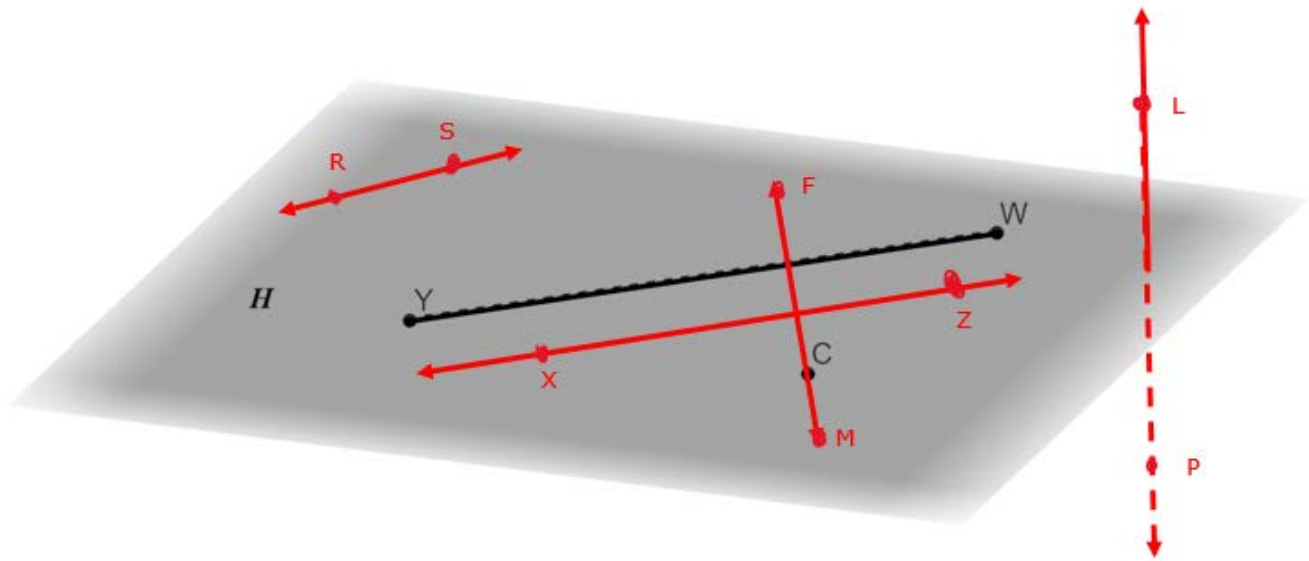
9. Identify two lines, if there are any, that do not intersect and lie on the same plane.

$\overleftrightarrow{WM}$ and $\overleftrightarrow{TL}$

10. Identify two lines, if there are any, that are perpendicular.

$\overleftrightarrow{DW}$ and $\overleftrightarrow{WM}$

Line segment YW and point C are shown on plane H .

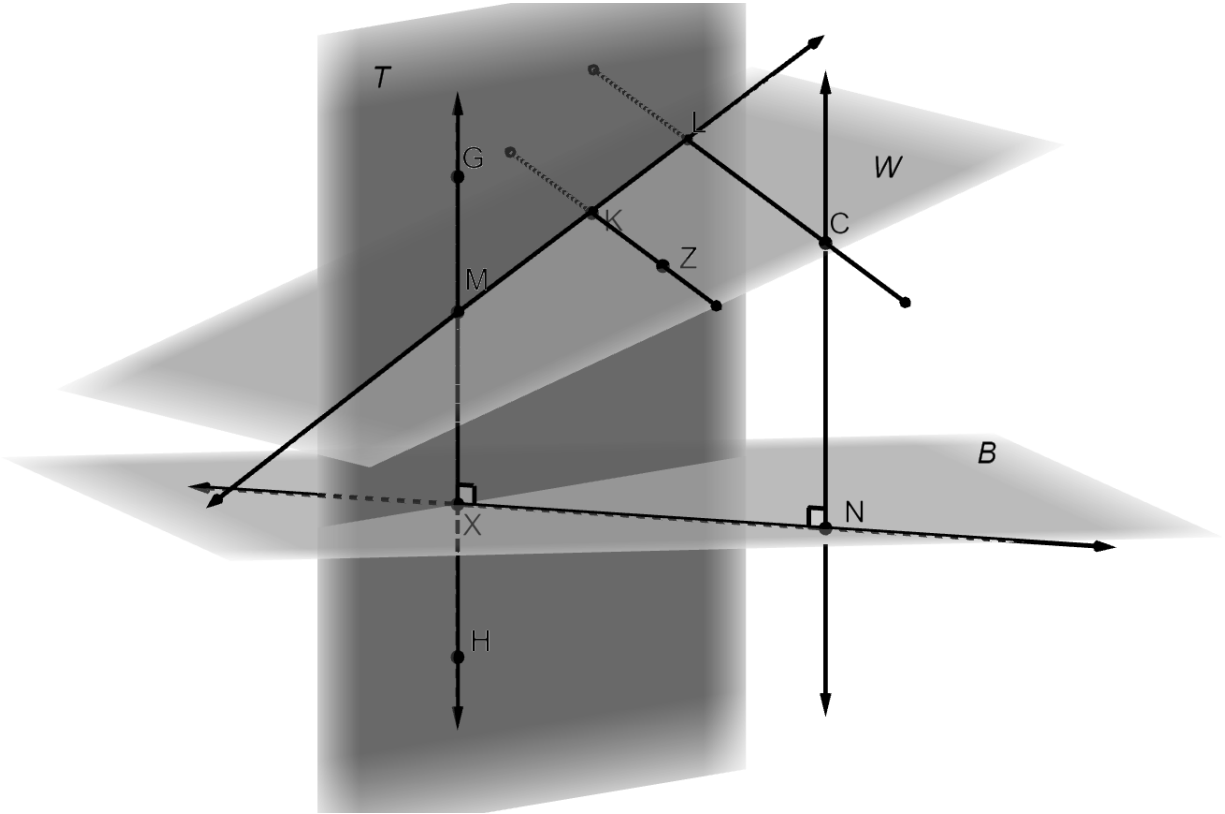


Draw an example of each geometric object on plane H . Label your drawing. If it is not possible to create a drawing then explain why it is not possible. **Note: Answers will vary. Sample answers are shown.**

	Geometric Object	Label	Explanation
1.	A line that lies in plane H .	$\overleftrightarrow{RS}$	
2.	A line that is skew to the $\overline{YW}$.	$\overleftrightarrow{LP}$	
3.	A line that is parallel to $\overline{YW}$ that does not go through point C .	$\overleftrightarrow{XZ}$	
4.	A line segment that is perpendicular to $\overline{YW}$ that goes through point C .	$\overline{FM}$	

5. Select all of the geometric objects in the drawing that are undefined.
☒ point C
☒ plane H
☐ line segment YW
☐ the skew line you drew
☐ the parallel line you drew
☐ the perpendicular line you drew
- Since point, line, and plane are undefined terms, point C and plane H are the correct answers.

Planes T , W , and B are shown.



List the line(s) that appear to lie in each plane. **Answers will vary.**

	Plane	Line(s)
6.	B	$\overleftrightarrow{XN}$
7.	T	$\overleftrightarrow{GX}$ and $\overleftrightarrow{ML}$
8.	W	$\overleftrightarrow{ML}$

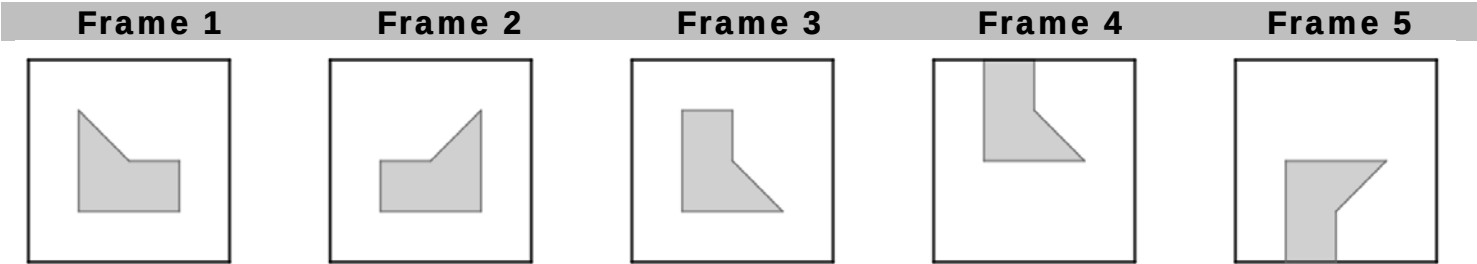
9. Name a pair of lines that are perpendicular.

$$\begin{aligned}\overleftrightarrow{XN} &\perp \overleftrightarrow{GX} \\ \overleftrightarrow{XN} &\perp \overleftrightarrow{CN}\end{aligned}$$

10. Name a pair of lines that are skew.

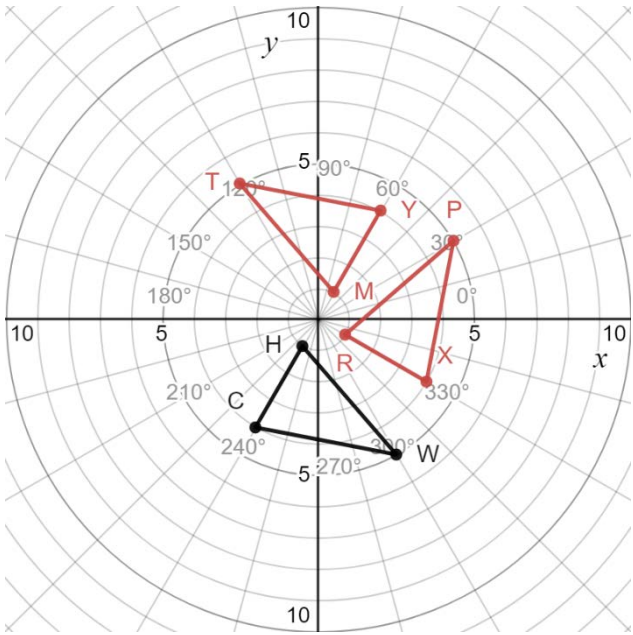
$$\overleftrightarrow{LM} \text{ and } \overleftrightarrow{XN}, \overleftrightarrow{LM} \text{ and } \overleftrightarrow{CN}$$

Five frames that feature a polygon in different positions are shown.



	Movement from...	Description
1.	Frame 1 to Frame 2	Reflection
2.	Frame 2 to Frame 3	Rotation
3.	Frame 3 to Frame 4	Translation
4.	Frame 4 to Frame 5	Reflection

Triangle *HCW* is shown on the circular graph.



5. Plot each of the points.

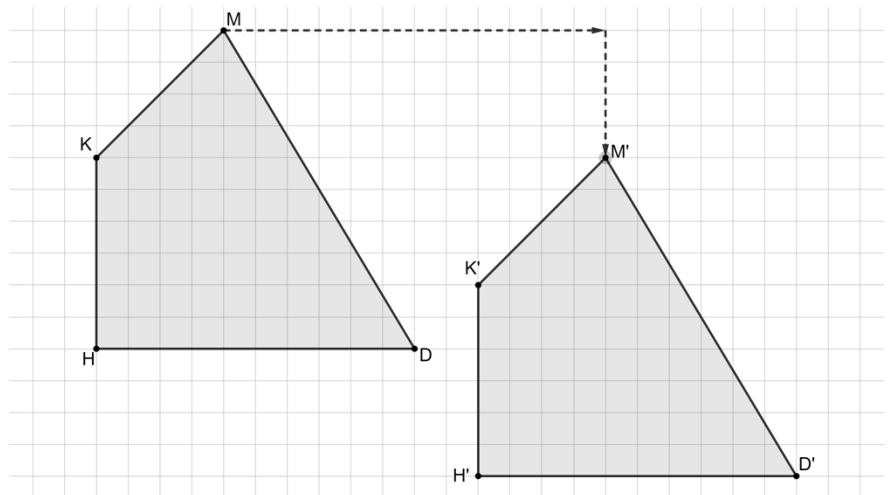
Point	Description
T	Plotted on the 120° line and on the same circle as W .
Y	Plotted on the 60° line and on the same circle as C .
M	Plotted on the 60° line and on the same circle as H .
P	Plotted on the 30° line and on the same circle as W .
X	Plotted on the 330° line and on the same circle as C .
R	Plotted on the 330° line and on the same circle as H .

6. Complete the statements about the triangles.

$\triangle TYM$ is a ☐ reflection ☒ rotation of $\triangle WCH$. $\triangle PXR$ is a ☐ reflection ☒ rotation of $\triangle WCH$.

A diagram of $MDHK$ and $M'D'H'K'$ is shown. Arrows are drawn from vertex M to vertex M' .

Complete the statements.



7. The arrows show a possible path of movement. This path is called a

- ☐ reflection.
☐ rotation.
☒ translation.

8. The same path ☒ exists ☐ does not exist for all points on a diagram.

9. The path is called ☐ isometry. ☒ a mapping.

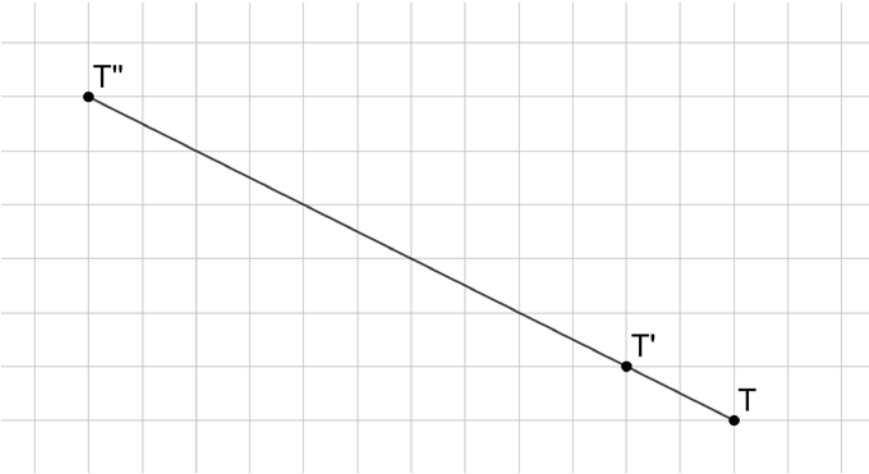
10. The transformation shown on the grid can be referred to as

or as ☐ an isometry ☐ a mapping ☒ a rigid motion because the transformation

- ☐ preserved distance.
☐ preserved angle measure.
☒ preserved angle measure and distance.

- ☒ an isometry
☐ a mapping
☐ a rigid motion

Points T , T' , and T'' are shown.



- 1. Determine the number of copies of $\overline{TT'}$ that will fit on $\overline{TT''}$. 6
- 2. Use the definition of dilation to explain why $\overline{TT''}$ is a dilation of $\overline{TT'}$.
 $\overline{TT''}$ is a proportional increase of $\overline{TT'}$. It is 6 times as long as $\overline{TT'}$.
- 3. Complete the statement.

Point

☐ T

☒ T'

☐ T''

 partitions

☐ $\overline{TT'}$

☐ $\overline{T'T''}$

☒ $\overline{TT''}$

 so that the ratio of

☐ $\frac{TT'}{T'T''}$

☒ $\frac{TT'}{TT''}$

 is

☐ $\frac{1}{5}$

☒ $\frac{1}{6}$

Use the Pythagorean theorem to complete the table.

	Segment	Length
4.	$\overline{TT'}$	$2^2 + 1^2 = c^2$ $5 = c^2$ $\sqrt{5} = c$
5.	$\overline{T'T''}$	$5^2 + 10^2 = c^2$ $125 = c^2$ $5\sqrt{5} = c$
6.	$\overline{TT''}$	$6^2 + 12^2 = c^2$ $180 = c^2$ $6\sqrt{5} = c$

7. Complete the statements.

Point T' is

☐ $\frac{1}{5}$

☒ $\frac{1}{6}$

 of the distance from T to T'' . The scale factor that can be used to

multiply

☐ $\overline{TT'}$

☐ $\overline{T'T''}$

☒ $\overline{TT''}$

 by to produce

☒ $\overline{TT'}$

☐ $\overline{T'T''}$

☐ $\overline{TT''}$

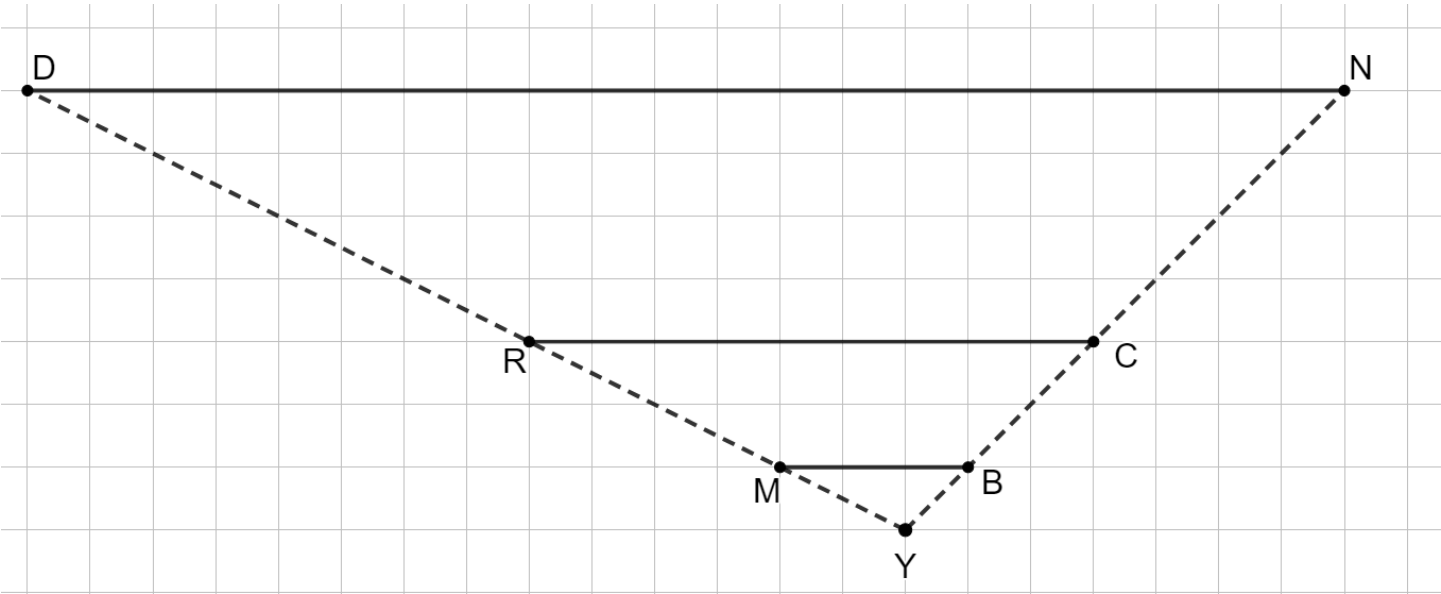
 is

☐ $\frac{1}{5}$

☒ $\frac{1}{6}$

.

Line segments MB , RC , and DN are shown where $\overline{MB}$ is dilated using point Y to create $\overline{RC}$ and $\overline{DN}$.



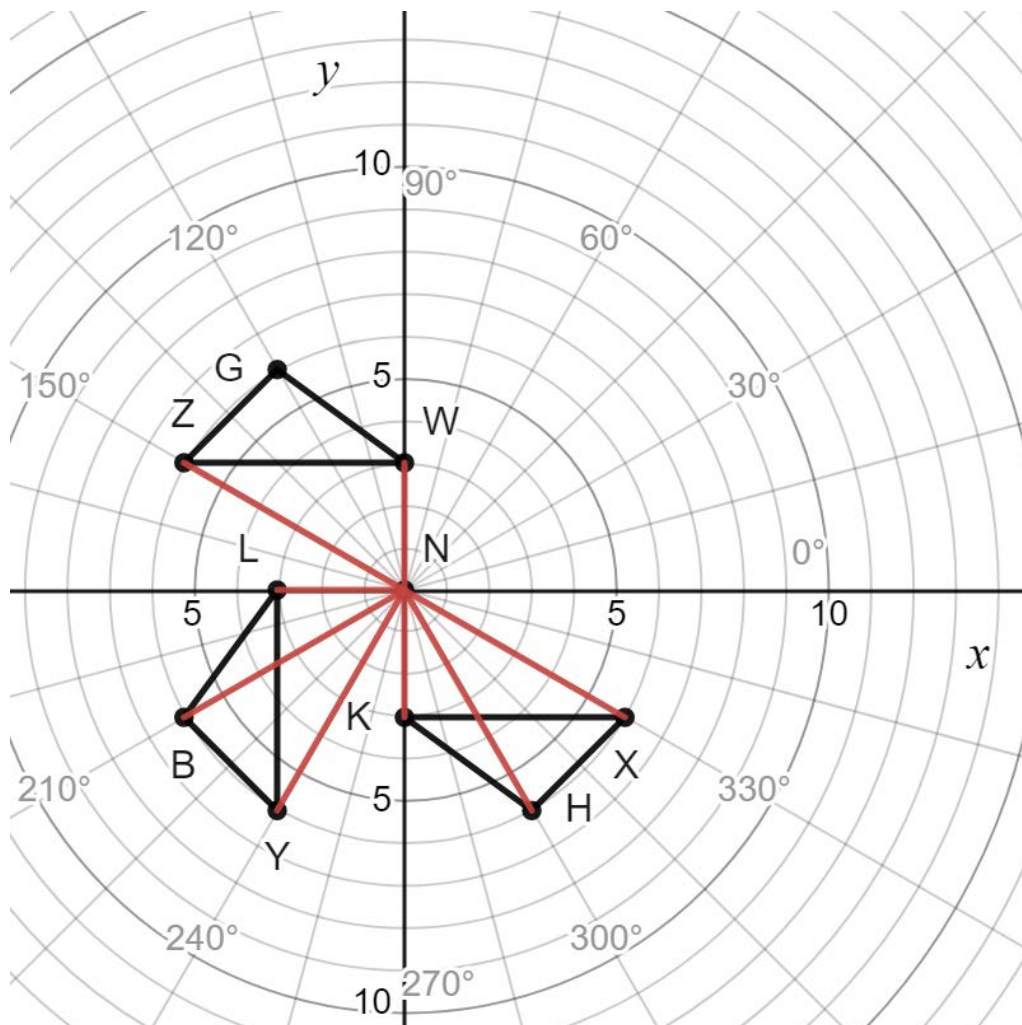
Determine each of the ratios.

	Ratio	Length
8.	$YM:YR$	$\sqrt{5}:3\sqrt{5}$ $1:3$
9.	$MB:RC$	$3:9$ $1:3$
10.	$MB:DN$	$3:21$ $1:7$

Geometry
Unit 1
Lesson 5 Practice

Name Answer Key
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Triangles GWZ , BLY , and HKX are shown on the circular grid.



Draw each of the angles on the grid.

1. $\angle LNW$
2. $\angle XNZ$
3. $\angle BNH$
4. $\angle XNY$
5. $\angle WNK$
6. $\angle YNZ$

7. Select all of the angles drawn on the grid whose measure is 90° .

- ☒ $\angle LNW$
- ☐ $\angle XNZ$
- ☒ $\angle BNH$
- ☒ $\angle XNY$
- ☐ $\angle WNK$
- ☒ $\angle YNZ$

8. Select all of the angles drawn on the grid whose measure is 180° .

- ☐ $\angle LNW$
- ☒ $\angle XNZ$
- ☐ $\angle BNH$
- ☐ $\angle XNY$
- ☒ $\angle WNK$
- ☐ $\angle YNZ$

Complete the statements.

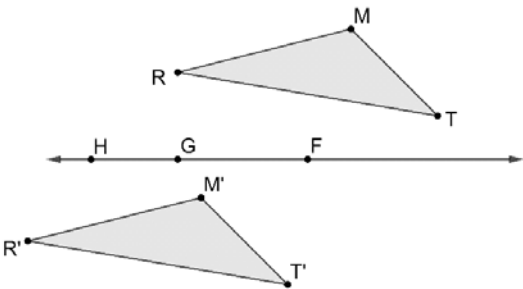
9. $\triangle GWZ$ is created by rotating ☐ $\triangle BLY$
☒ $\triangle HKX$ 180° counterclockwise about point N .

10. $\triangle BLY$ is created by rotating ☐ $\triangle HKX$
☒ $\triangle GWZ$ 90° counterclockwise about point N .

Triangles RMT and $R'M'T'$ are shown.

1. Complete the table to describe the single transformation that was used to create $R'M'T'$.

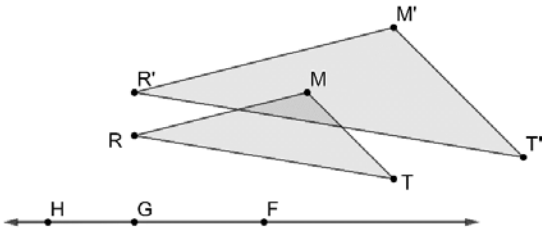
Type	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☒ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☒ Down ☒ Left ☐ Right ☐ Up	



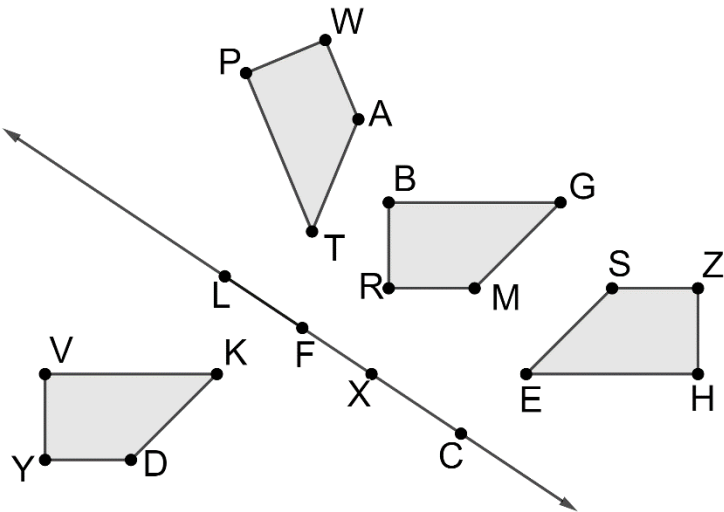
Triangles RMT and $R'M'T'$ are shown.

2. Complete the table to describe the single transformation that was used to create $R'M'T'$.

Type	Center, Line, or Movement	Degrees or Scale Factor
☒ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☒ Center of dilation is G . ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	



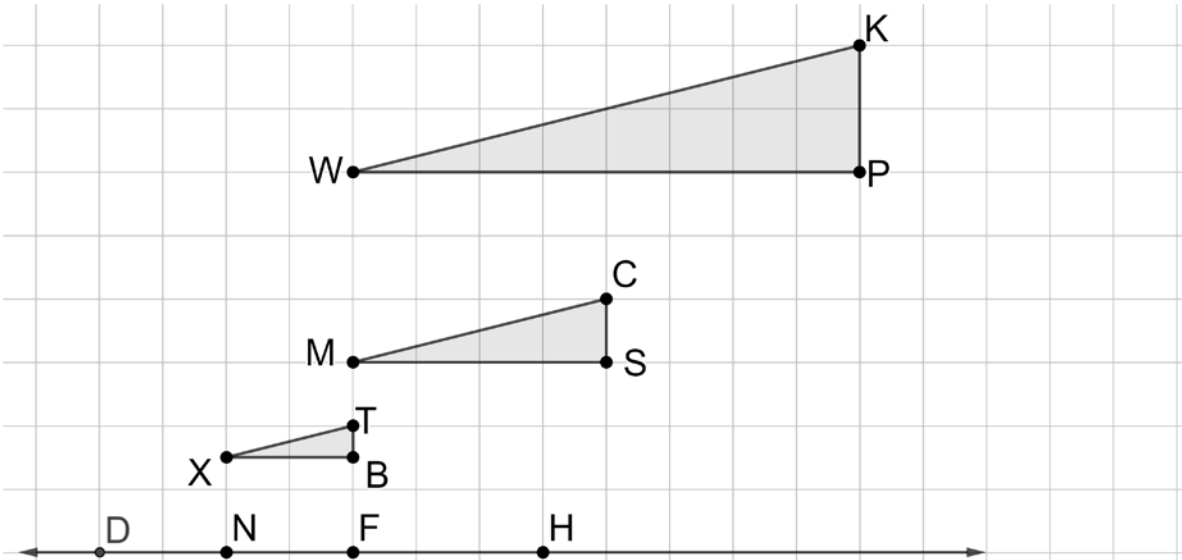
The diagram shows three different single transformations of quadrilateral $VKDY$.



Complete the table.

	Quadrilateral	Type of Transformation
3.	$PTAW$	Reflection
4.	$BGMR$	Translation
5.	$HESZ$	Rotation

The diagram shows two different dilations of $\triangle MCS$.



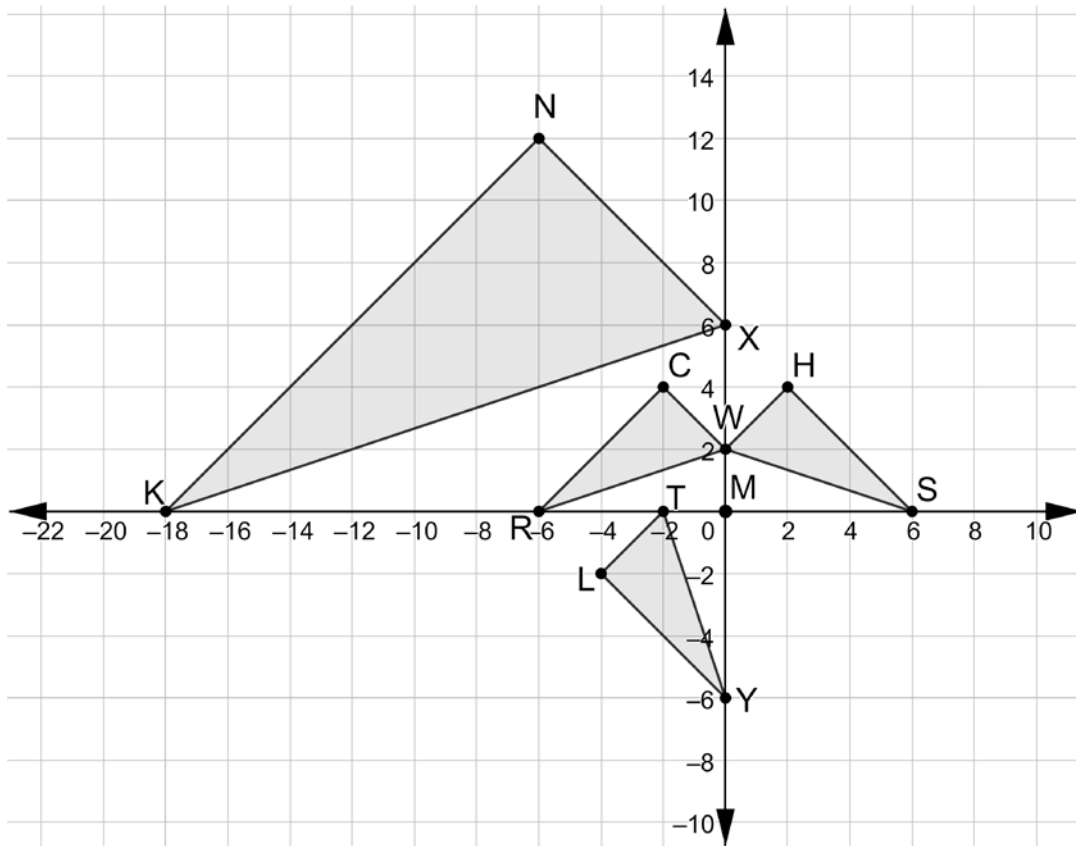
Complete the statements.

- Triangle XTB is a dilation of $\triangle MCS$ with a center of dilation at point D and a scale factor of $\frac{1}{2}$.
- Triangle WKP is a dilation of $\triangle MCS$ with a center of dilation at point F and a scale factor of 2.

Geometry
Unit 1
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

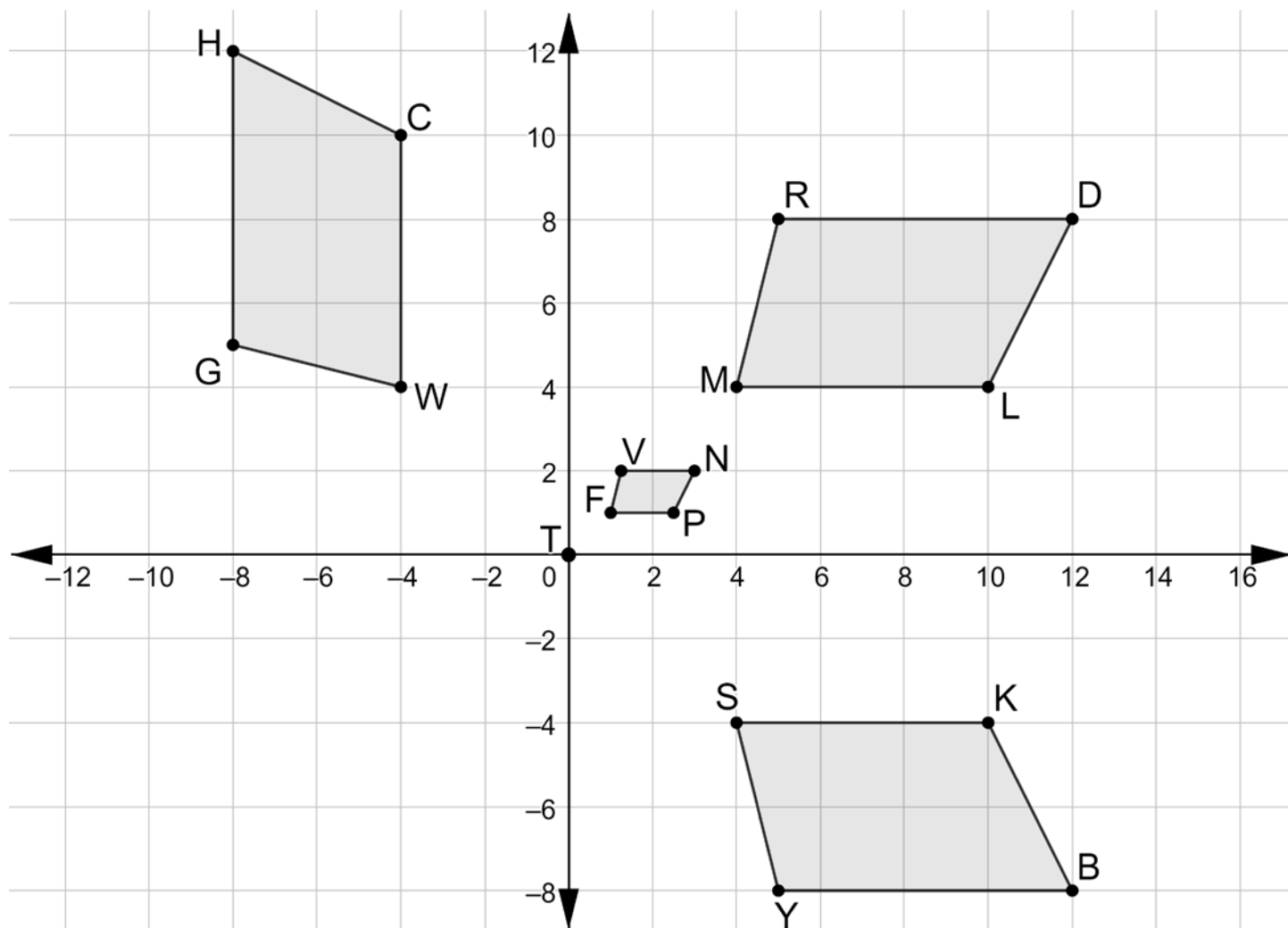
Four triangles are shown on the coordinate grid. $\triangle WSH$, $\triangle TYL$, and $\triangle XKN$ are created using a single transformation of $\triangle WRC$.



Complete each statement using an angle or a segment from $\triangle WRC$.

1. $\overline{HW} \cong \overline{CW}$
2. $\overline{SW} \cong \overline{RW}$
3. $\overline{LY} \cong \overline{CR}$
4. $\angle TYL \cong \angle WRC$
5. $\overline{NK} \sim \overline{CR}$
6. $\overline{XK} \sim \overline{WR}$

Four quadrilaterals are shown on the coordinate grid. Quadrilaterals $VNPF$, $GHCW$, and $YBKS$ are created using a single transformation of quadrilateral $RDLM$.



Complete each statement using an angle or segment from quadrilateral $RDLM$.

7. $\overline{CW} \cong \overline{LM}$
8. $\angle YSK \cong \angle RML$
9. $\overline{NV} \sim \overline{DR}$
10. $\angle CHG \cong \angle LDR$

Geometry
Unit 1
Lesson 8 Practice

Name Answer Key
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Line segment KN is shown where $KR = 16x - 3$, $RW = WN$, $WN = 9x - 3$, $BR = \frac{1}{2}RW$, and $KN = 161$.



1. Find the value of x .

$$\begin{aligned}16x - 3 + 9x - 3 + 9x - 3 &= 161 \\34x - 9 &= 161 \\34x &= 170 \\x &= 5\end{aligned}$$

2. Find the length of $\overline{RW}$.

$$9 \cdot 5 - 3 = 45 - 3 = 42$$

3. Find the length of $\overline{KW}$.

$$\begin{aligned}16 \cdot 5 - 3 &= 80 - 3 = 77 \\77 + 42 &= 119 \\119\end{aligned}$$

4. Find the length of $\overline{BR}$.

$$\frac{42}{2} = 21$$

5. Use a compass and straightedge to make a copy of $\overline{HT}$.



T , M , W , and H are four collinear points that exist such that $\overline{TW}$ and $\overline{MH}$ are congruent. $TW = 5y + 10$ and $MH = 9y - 46$. Four different students draw a diagram of the description. Each student's drawing is shown.

Name	Drawing
Miquel	
Harper	
Camila	
Liam	

6. Determine the value of y .

$$\begin{aligned}
 5y + 10 &= 9y - 46 \\
 56 &= 4y \\
 14 &= y
 \end{aligned}$$

7. Use Miquel's drawing to determine the length of $\overline{WM}$ if $\frac{7}{3}WM = MH$.

$$\begin{aligned}
 MH &= 9y - 46 = 9 \cdot 14 - 46 = 80 \\
 \frac{7}{3}WM &= MH; \quad \frac{7}{3}WM = MH; \quad \frac{7}{3}WM = 80 \\
 WM &= 80 \cdot \frac{3}{7} = \frac{240}{7} = 34\frac{2}{7}
 \end{aligned}$$

8. Use Harper's drawing to determine the length of $\overline{MW}$ if $\frac{3}{5}TM = MW$.

$$\begin{aligned}
 TM + MW &= TW; \quad TM + \frac{3}{5}TM = 5 \cdot 14 + 10; \quad \frac{8}{5}TM = 80 \\
 TM &= 50; \quad MW = 80 - 50 = 30
 \end{aligned}$$

9. Use Camila's drawing to determine the length of $\overline{MW}$ if $TM + MW + WH = 130$.

$$80 + WH = 130; \quad WH = 50; \quad 130 - 100 = 30 = MW$$

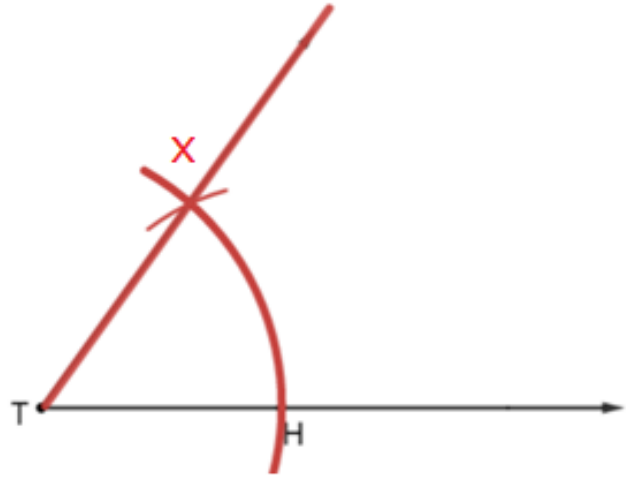
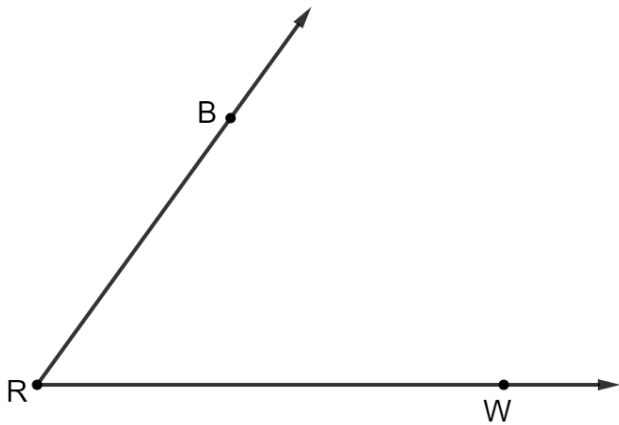
10. Use Liam's drawing to determine the length of $\overline{WM}$ if $TH = 190\frac{1}{3}$.

$$\begin{aligned}
 80 + x + 80 &= 190\frac{1}{3} \\
 x &= 30\frac{1}{3}
 \end{aligned}$$

Geometry
Unit 1
Lesson 9 Practice

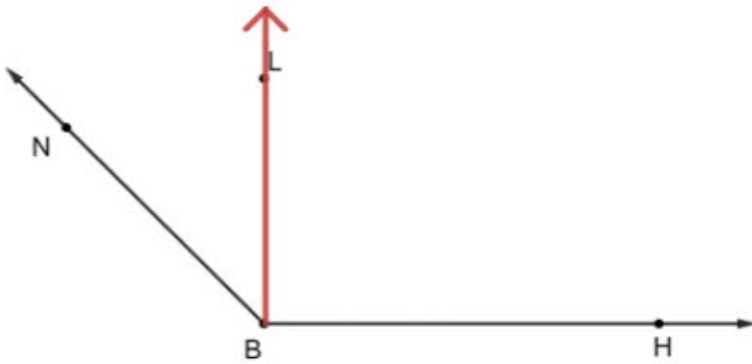
Name Answer Key
Date _____
Period _____

Angle BRW and ray TH are shown.



1. Use $\overrightarrow{TH}$ to create $\angle HTX$, a copy of $\angle BRW$.

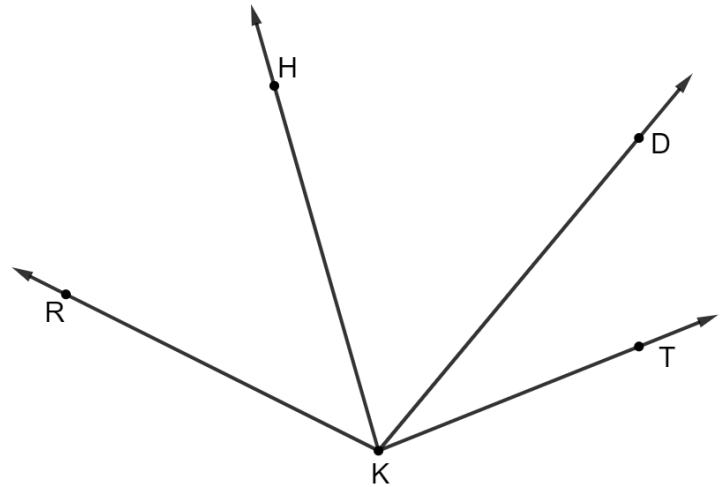
Point L lies in the interior of angle NBH as shown.



2. Draw ray BL .
3. Complete the statement.

Since point L is in the interior of $\angle NBH$ then $m\angle NBL + m\angle LBH = m\angle NBH$.

Angles RKH , HKD , and DKT are shown where $m\angle RKH = (4y)^\circ$, $m\angle DKH = (6y - 16)^\circ$, $m\angle DKT = (2y + 4)^\circ$, and $m\angle RKT = 132^\circ$.



4. Determine the value of y .
 $4y + 6y - 16 + 2y + 4 = 132$
 $12y - 12 = 132$
 $12y = 144$
 $y = 12$

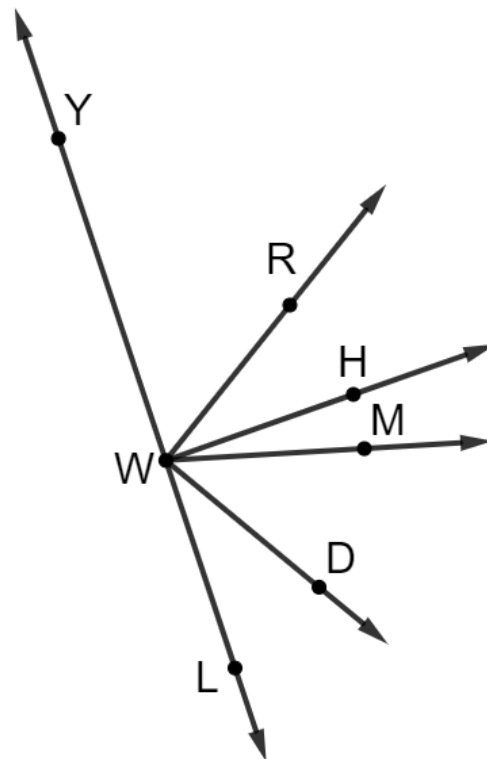
5. Determine $m\angle HKT$.
 $6y - 16 + 2y + 4$
 $8y - 12$
 $8 \cdot 12 - 12$
 84°

6. Determine $m\angle DKR$.
 $6y - 16 + 4y$
 $10y - 16$
 $10 \cdot 12 - 16$
 104°

Points R , H , M , and D lie in the interior of straight angle YWL such that $m\angle YWM = 105^\circ$, $m\angle MWD = 43^\circ$, $m\angle HWD = 59^\circ$, and $m\angle RWH = 2m\angle HWM$.

Determine each.

7. $m\angle YWD$
 $105^\circ + 43^\circ = 148^\circ$
8. $m\angle LWD$
 $180^\circ - 148^\circ = 32^\circ$
9. $m\angle HWM$
 $59^\circ - 43^\circ = 16^\circ$
10. $m\angle HWR$
 $2 \cdot 16^\circ = 32^\circ$



Geometry
Unit 1
Lesson 10 Practice

Name **Answer Key** _____
Date _____
Period _____

$\overline{WT}$, $\overline{CM}$, $\overleftrightarrow{XK}$, $\overleftrightarrow{LG}$, and $\overleftrightarrow{LP}$ are shown.

Use patty paper or fold this paper to determine each.

1. Which line or segment, if any, bisects $\overline{CM}$.

$\overleftrightarrow{XK}$

2. Which line or segment, if any, bisects $\overline{WT}$.

$\overline{CM}$

3. Which line or segment, if any, bisects $\overline{RD}$.

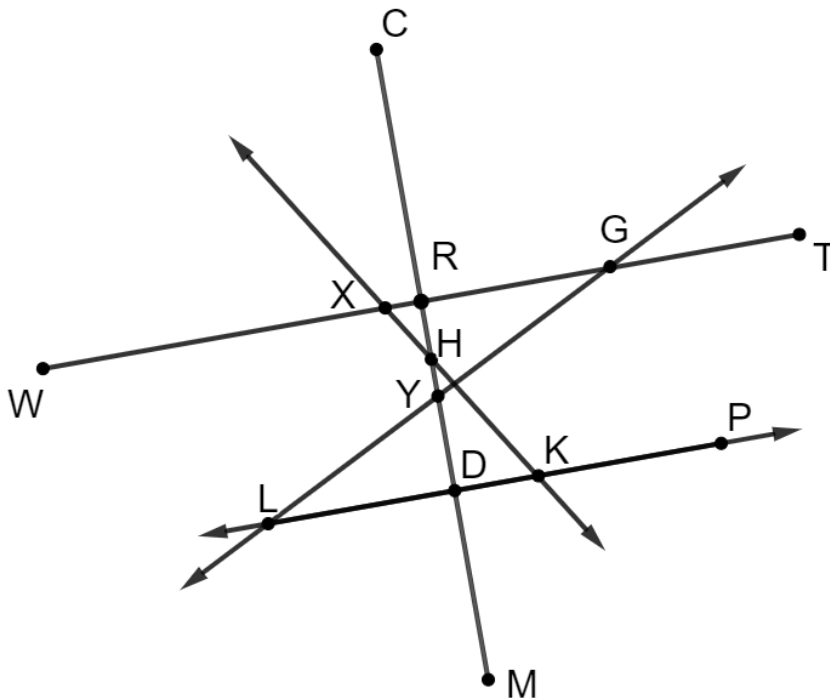
$\overleftrightarrow{LG}$

4. Which line or segment, if any, bisects $\overline{RT}$.

$\overleftrightarrow{LG}$

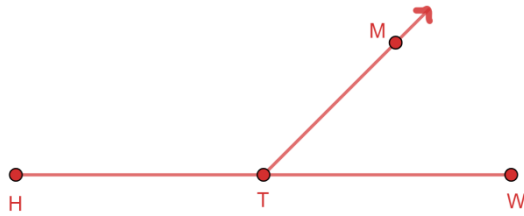
5. Complete the statement.

$\overline{CM}$ is a perpendicular bisector of $\overline{WT}$.



$\overrightarrow{TM}$ bisects $\overline{HW}$ at point T .

6. Draw a diagram of the description.



7. Determine the value of k if $HT = 5k + 12$ and $TW = 10k - 23$.

$$5k + 12 = 10k - 23$$

$$12 = 5k - 23$$

$$35 = 5k$$

$$7 = k$$

8. What is the length of $\overline{HW}$?

$$HW = 2(HT)$$

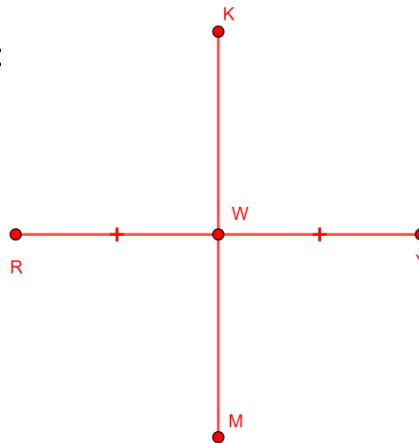
$$= 2(5 \cdot 7 + 12)$$

$$= 2(47)$$

$$= 94$$

$\overline{KM}$ is a perpendicular bisector of $\overline{RY}$ where point W is the midpoint.

9. Draw a diagram of the description



10. Determine the value of x if $RY = 11x + 46$ and $RW = 3x + 58$.

$$11x + 46 = 2(3x + 58)$$

$$11x + 46 = 6x + 116$$

$$5x + 46 = 116$$

$$5x = 70$$

$$x = 14$$

Geometry
Unit 1
Lesson 11 Practice

Name **Answer Key**
Date
Period

Steps to construct the perpendicular bisector of $\overline{HM}$ are shown out of order.

Step	Description
A	Repeat Steps 1 and 2 to create an arc centered at point M .
B	Label the point of intersection of $\overline{TW}$ and $\overline{HM}$ as R .
C	Without changing the width of the compass, draw an arc above and below the line centered at point H .
D	Draw $\overline{TW}$.
E	Set the compass point at point H and open the compass to a width approximately two-thirds the length of $\overline{HM}$.
F	Mark the two points of intersections of the arcs as T and W .

Use the letter of the steps to correctly reorder the steps for constructing a perpendicular bisector.

	Letter
1.	<i>E</i>
2.	<i>C</i>
3.	<i>A</i>
4.	<i>F</i>
5.	<i>D</i>
6.	<i>B</i>

$\overline{YK}$ is a perpendicular bisector of $\overline{HB}$. $HY = 7x + 13$, $HW = 8y - 15$, $WB = 3y + 21$, and $YB = 12x - 12$.

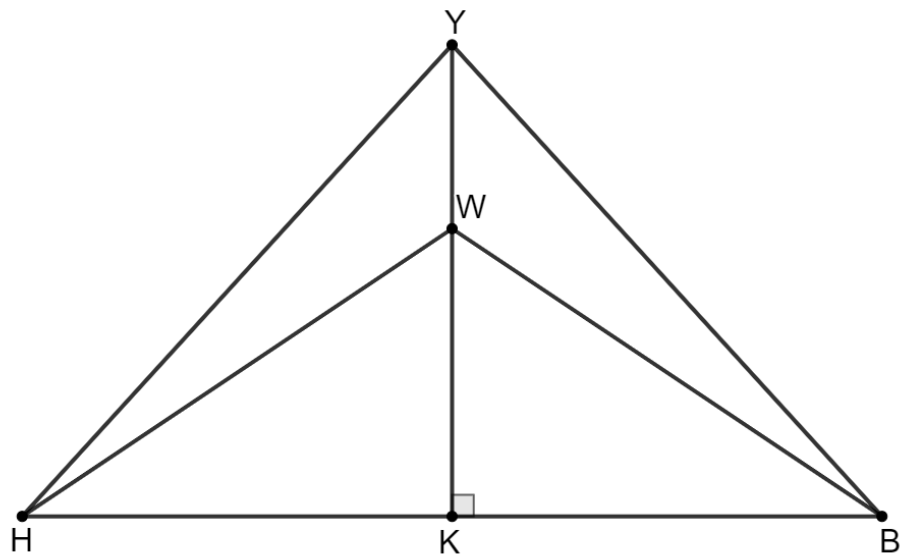
7. Determine the value of x .

$$7x + 13 = 12x - 12$$

$$13 = 5x - 12$$

$$25 = 5x$$

$$5 = x$$



8. Determine the length of $\overline{HY}$.

$$7(5) + 13 = 48$$

9. Determine the value of y .

$$8y - 15 = 3y + 21$$

$$5y - 15 = 21$$

$$5y = 36$$

$$y = 7.2$$

10. Determine the length of $\overline{WB}$.

$$3(7.2) + 21 = 42.6$$

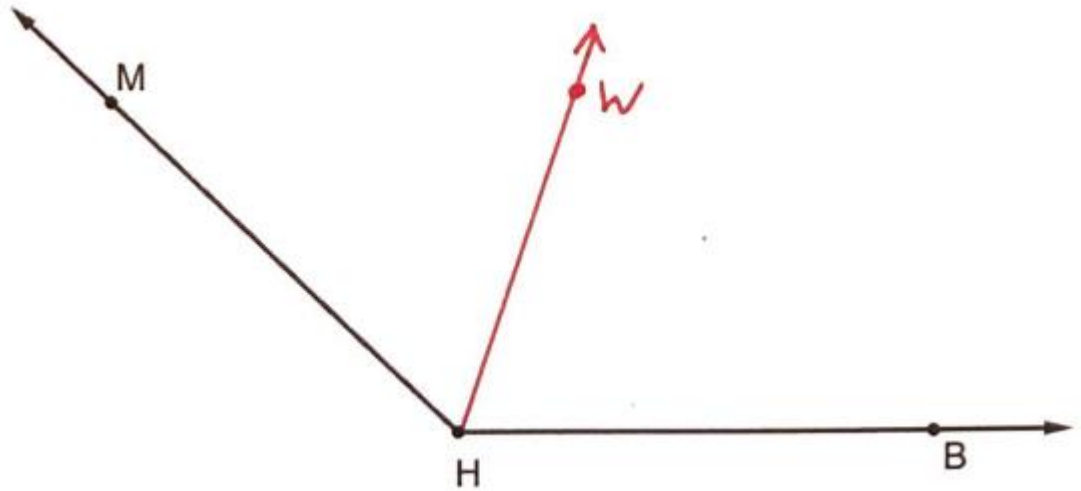
Geometry
Unit 1
Lesson 12 Practice

Name Answer Key
Date _____
Period _____

$\angle MHB$ is shown.

1. Fold the paper so that $\overrightarrow{HM}$ and $\overrightarrow{HB}$ lie on each other.

2. Use the fold to draw $\overrightarrow{HW}$.



3. Complete the statement.

$\overrightarrow{HW}$ bisects angle MHB so that $\angle$ MHW $\cong$ $\angle$ BHW .

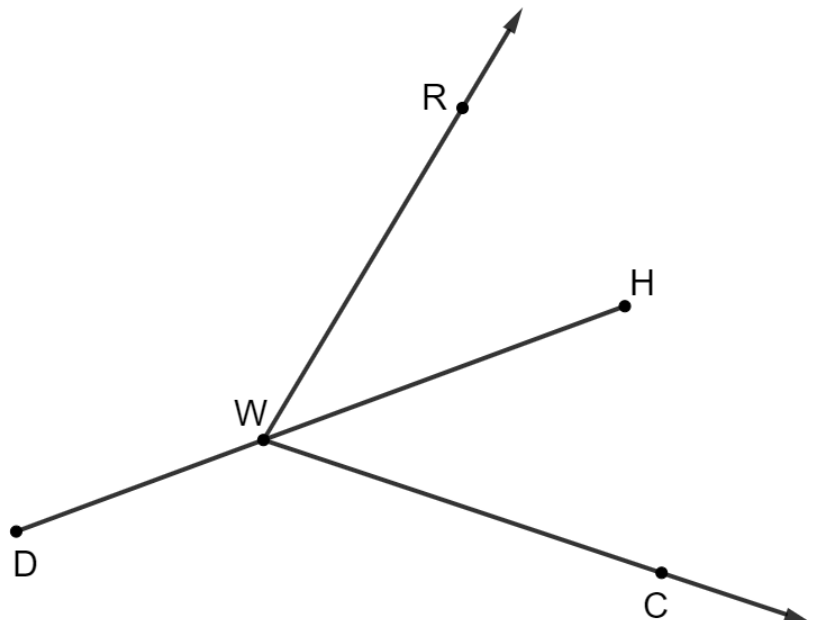
Line segment DH bisects $\angle RWC$ such that $m\angle RWH = (7k + 16.3)^\circ$ and $m\angle CWH = (13k - 2.9)^\circ$.

4. Determine the value of k .

$$\begin{aligned} 7k + 16.3 &= 13k - 2.9 \\ 16.3 &= 6k - 2.9 \\ 19.2 &= 6k \\ k &= 3.2 \end{aligned}$$

5. Determine $m\angle RWH$.

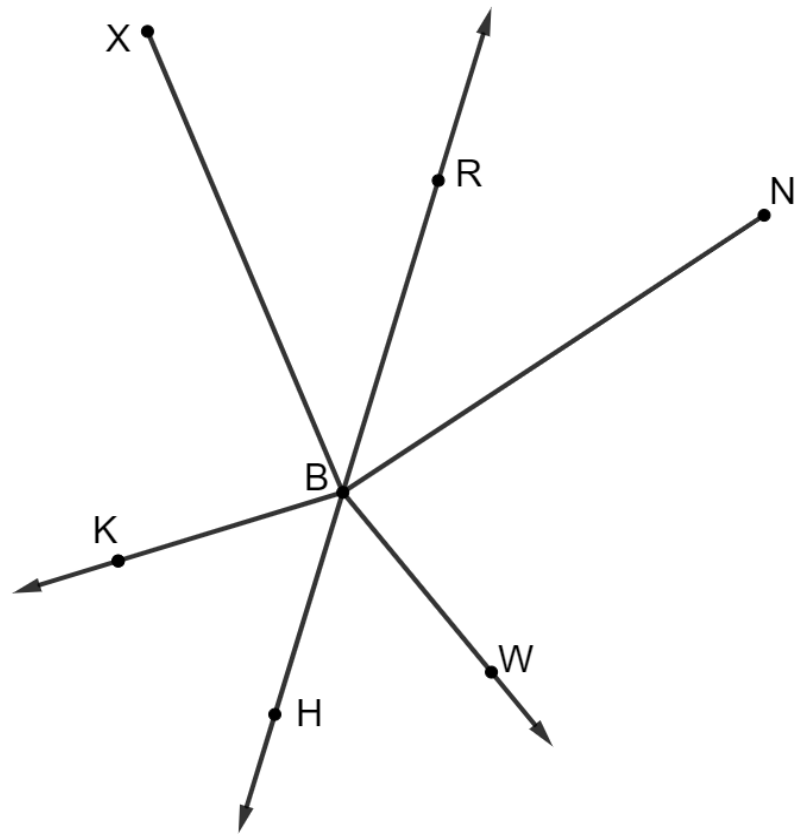
$$\begin{aligned} m\angle RWH &= 7(3.2) + 16.3 \\ &= 22.4 + 16.3 = 38.7^\circ \end{aligned}$$



6. Determine $m\angle CWD$.

$$m\angle CWD = 180 - 38.7 = 141.3^\circ$$

Line HR bisects $\angle XBN$ and $\angle KBW$ such that $m\angle XBR = (2x + 7.5)^\circ$, $m\angle KBH = (3y - 7)^\circ$, $m\angle WBH = (7y - 91)^\circ$, and $m\angle XBN = (5x - 1)^\circ$.



7. Determine the value of x .

$$2 \cdot m\angle XBR = m\angle XBN$$

$$2(2x + 7.5) = 5x - 1$$

$$4x + 15 = 5x - 1$$

$$15 = x - 1$$

$$16 = x$$

8. Determine the value of y .

$$3y - 7 = 7y - 91$$

$$-7 = 4y - 91$$

$$84 = 4y$$

$$21 = y$$

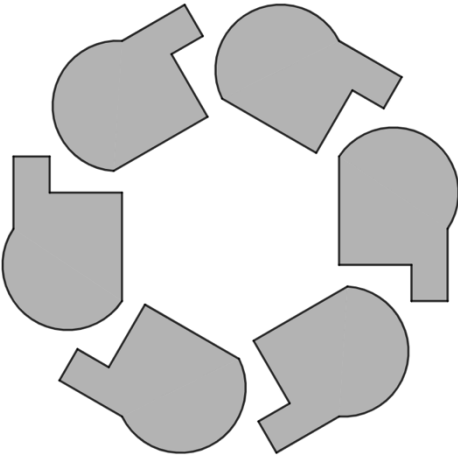
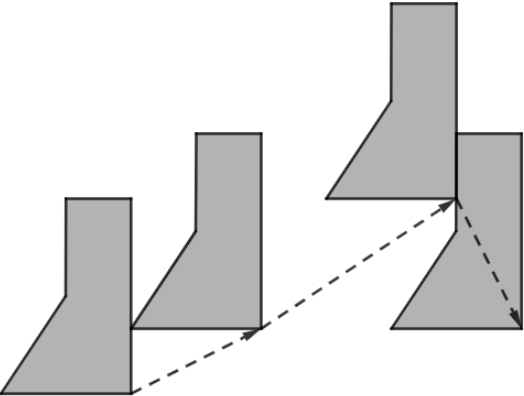
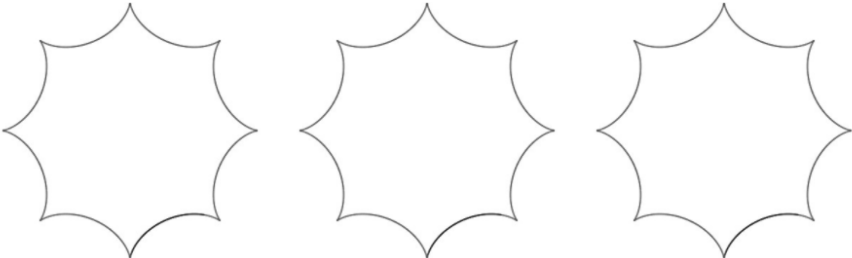
9. Determine $m\angle RBN$.

$$m\angle RBN = m\angle XBR = 2x + 7.5 = 2(16) + 7.5 = 32 + 7.5 = 39.5^\circ$$

10. Determine $m\angle KBW$.

$$m\angle KBW = 2(m\angle KBH) = 2(3 \cdot 21 - 7) = 2(63 - 7) = 2(56) = 112^\circ$$

For each set of images name the type(s) of symmetry the image represents.

	Set of Images	Symmetry
1.		Rotational
2.		Reflectional
3.		Translational
4.		Rotational, Reflectional, and Translational

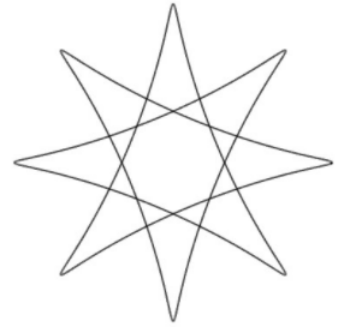
A figure is shown.

5. How many lines of symmetry does the figure have?

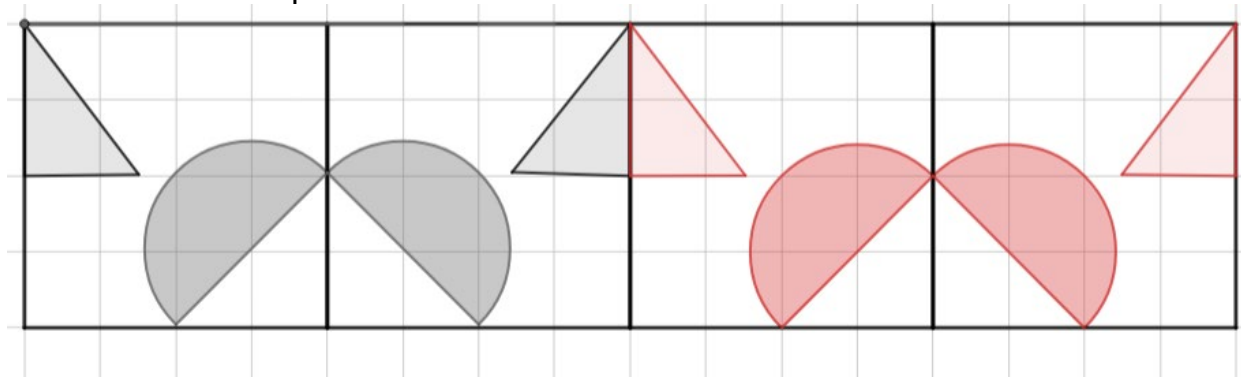
8

6. What is the order of rotational symmetry for the figure?

8th order



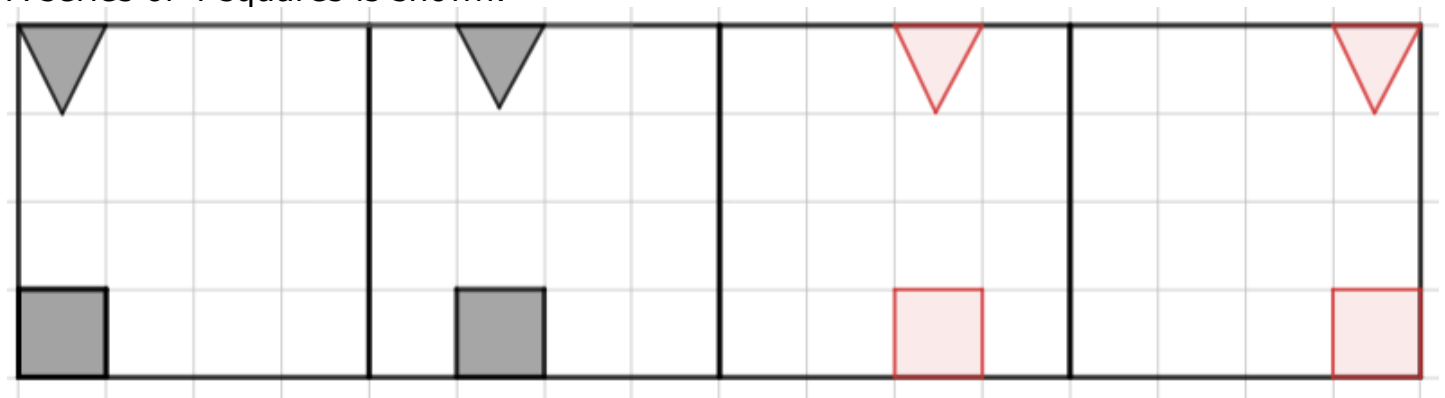
A series of four squares is shown.



7. Use the pattern in the first two squares to complete the pattern in the last two squares.
8. What type of symmetry is shown in this series of squares?

Reflectional

A series of 4 squares is shown.



9. Use the pattern in the first two squares to complete the pattern in the last two squares.
10. What type of symmetry is shown in this series of squares?

Translational